

AAA -- Authentication -- Guided Notes ANSWER KEY

Unit tp-6.5 | CompTIA Network+ N10-009 Obj. FC0-U71 6.1 | N10-008 FC0-U71 6.1

For instructor use / distribute after students complete the notes.

Question 1

AAA in security stands for _____, _____, and _____. Authentication answers: 'Who are _____?' Authorization answers: 'What are you _____ to do?' Accounting answers: 'What did you _____?'

Answer: Authentication; authorization; accounting; you; allowed; do.

Question 2

Authentication verifies that a user is who they claim to be. The three authentication factors are: something you _____ (password, PIN), something you _____ (smart card, security key, phone), and something you _____ (fingerprint, face, retina). Using two or more factors is called _____-factor authentication (MFA).

Answer: Know; have; are; multi.

Question 3

Passwords are the most common authentication factor but also the weakest when they are _____, _____, or _____. Password best practices include: minimum _____ characters, use of upper/lower/numbers/symbols, no _____ personal info (names, dates), and no password _____ across multiple sites.

Answer: Short; simple; reused; 12+ (8+); obvious; reuse.

Question 4

A _____ attack uses a list of common words and variations to crack passwords. A _____ force attack tries every possible combination. A _____ attack uses precomputed hash tables to find plaintext passwords. Account _____ after N failed attempts defeats all three approaches by limiting guesses.

Answer: Dictionary; brute; rainbow table; logout.

Question 5

Multi-factor authentication (MFA) requires a second form of verification in addition to a password. Common second factors include _____ codes sent via SMS, _____ app codes (TOTP), _____ security keys (FIDO2/WebAuthn), and biometric scans. SMS-based MFA is the _____ secure option because SIM swapping attacks can intercept codes.

Answer: One-time (OTP); authenticator; hardware; least.

Question 6

Single Sign-On (SSO) allows users to _____ once and gain access to multiple applications without re-entering credentials. SSO relies on a trusted _____ provider (IdP) that issues authentication

_____ to authorized applications. Examples include Microsoft Entra (formerly Azure AD), Google Workspace, and Okta.

Answer: Log in (authenticate); identity; tokens.

Question 7

Phishing attacks target authentication by tricking users into entering credentials on _____ websites that look real. Spear _____ is a targeted phishing attack aimed at a specific individual or organization. Phishing is the most common initial vector in _____ breaches. FIDO2 hardware keys are phishing-_____ because the key verifies the site domain before authenticating.

Answer: Fake (fraudulent); phishing; data; resistant.

Question 8 -- Real World Application

REAL WORLD: A company requires employees to use a 6-digit PIN as their only authentication factor for the internal HR system containing salary and personal data for all 500 employees. Describe two specific weaknesses of this authentication scheme and recommend a stronger implementation, explaining why each improvement helps.

Answer: Weakness 1 -- Single factor (no MFA): a 6-digit PIN is a 'something you know' factor only. If an attacker learns the PIN (via shoulder surfing, keylogging, phishing, or a database breach), they gain immediate full access with no second barrier. Recommendation: add a second factor -- require an authenticator app TOTP code (e.g., Microsoft Authenticator). Even if a PIN is stolen, the attacker also needs the employee phone to complete login. This eliminates the single-point-of-failure problem.

Weakness 2 -- Low entropy ($10^6 = 1$ million combinations): a 6-digit PIN can be brute-forced in seconds with no rate limiting. If the system does not lock accounts after failed attempts, an automated script can try all 1,000,000 combinations quickly. Recommendation: enforce account lockout after 5 failed attempts and require a minimum 12-character password with complexity requirements instead of a 6-digit PIN. This raises the brute force cost to impractical levels.